Baker College | 115 YEARS OF IMPACT

Precision Hand Tools

Baker College Engineering

Brad Peirson
College of Information Technology and Engineering

Agenda

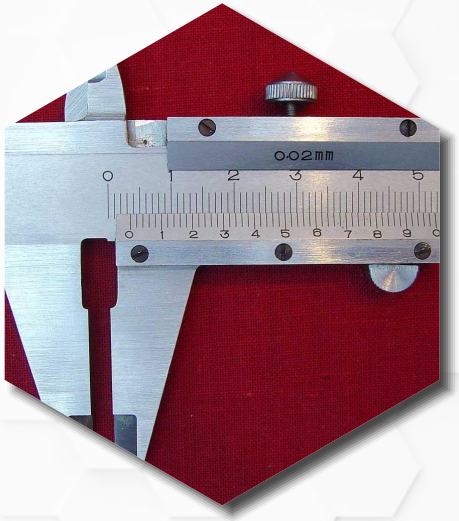
Vernier Instruments

Micrometers

Vernier Instruments

Vernier Instruments Applications

- Vernier is a type of scale
- Used on **many** different measurement devices
  - Including some micrometers



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Transition to Dial Tools

Most vernier instruments were replaced by dial instruments *LONG* ago...

Shift to Digital Calipers

...so long ago that most dial instruments have been replaced by digital instruments.

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Why Use Dial Instruments?

When possible, you should still shoot for dial instruments...why?

Reliability of Dial Calipers

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Batteries die–dial calipers will always be ready to use when you need them.

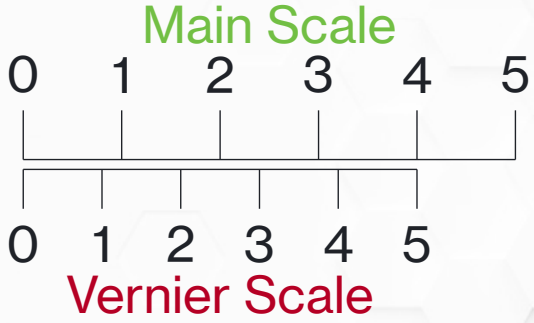
Vernier Instruments

Reading a Vernier Scale

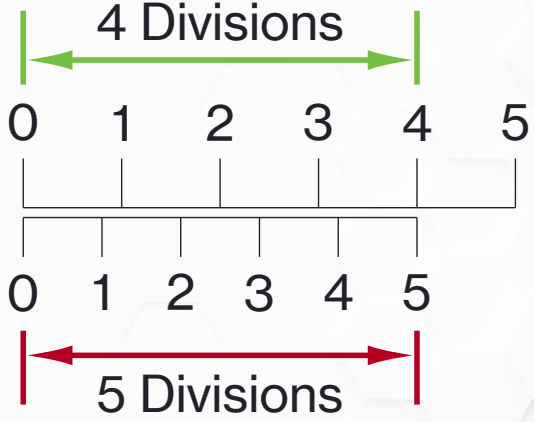
Scale Subdivisions Concept

- A **vernier scale** is a way to subdivide a larger scale
- The vernier scale works by measuring a percentage of the larger scale
- A distance on the vernier scale is one unit less than the same distance on the larger scale
- Huh?

5-Division Scale (1 of 6)

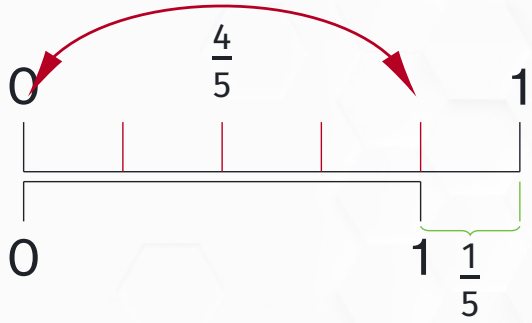


5-Division Scale (2 of 6)



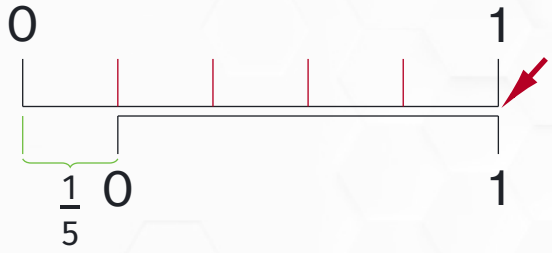
If the main scale has 5 divisions, the Vernier scale *also* have 5 divisions but will *fit them* in the space of 4 divisions of the main scale.

5-Division Scale (3 of 6)



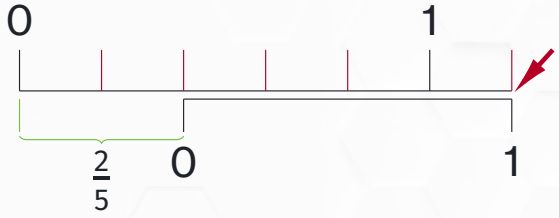
Imagine the same scale *really* zoomed in so we are looking at just the first marks–the first Vernier mark is *still* 1/5 short of the first main mark.

5-Division Scale (4 of 6)



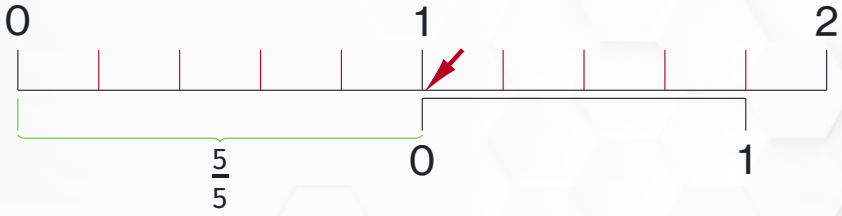
As we start to slide the scale the percent difference is what we're measuring. In this example, when we slide the Vernier so the 0 lines up with the first main division the measurement is 1/5.

5-Division Scale (5 of 6)



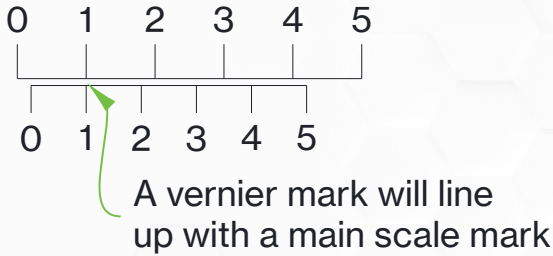
When we line up the Vernier zero with the *second* main mark the difference is 2/5.

5-Division Scale (6 of 6)



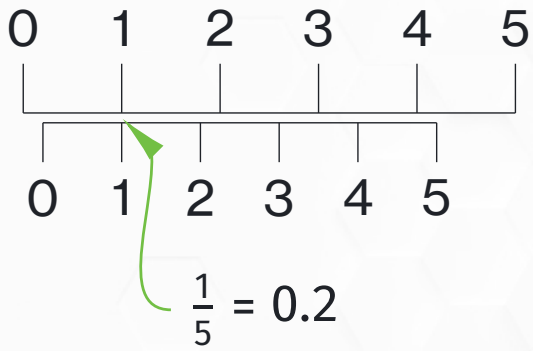
When the Vernier fully lines up with the major division, the difference is 5/5 or 1.

Reading Alignments (1 of 3)



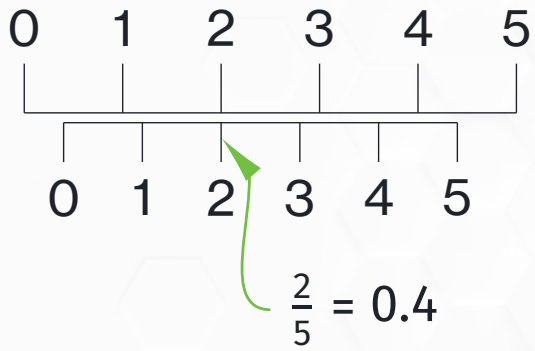
Let's zoom back out–when we move the Vernier to make a measurement one Vernier mark will line up with one mark on the main scale. We read the scale by looking at which mark lines up.

Reading Alignments (2 of 3)



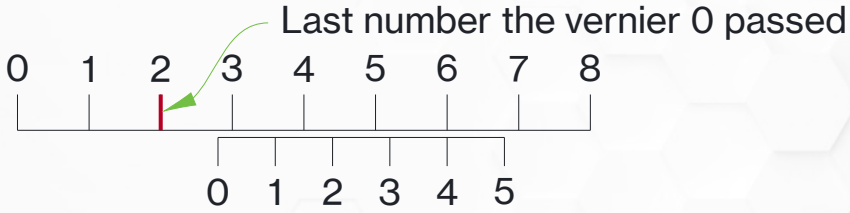
The mark that lines up tells us the fraction–mark 1 lines up in this diagram, the fraction is 1/5.

Reading Alignments (3 of 3)



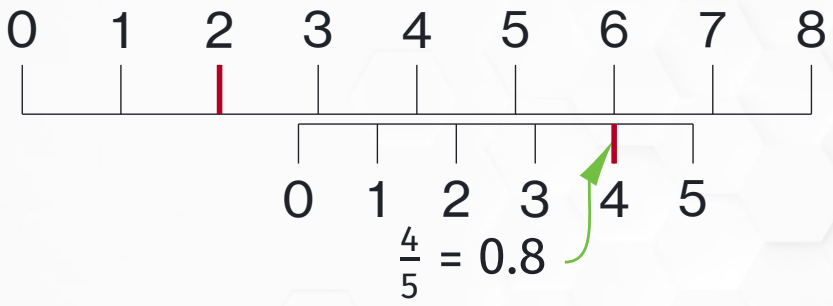
When the scale moves far enough to line up the second mark, the measurement is 2/5.

Offset Main Scale (1 of 2)



If the Vernier 0 hasn't passed a main mark we ignore the main scale. But if it *passes* a main mark we need to count that main mark first.

Offset Main Scale (2 of 2)



$2 + 0.8 = 2.8$

After counting the number on the main scale, *then* we look for the Vernier mark that lines up to get the total measurement.

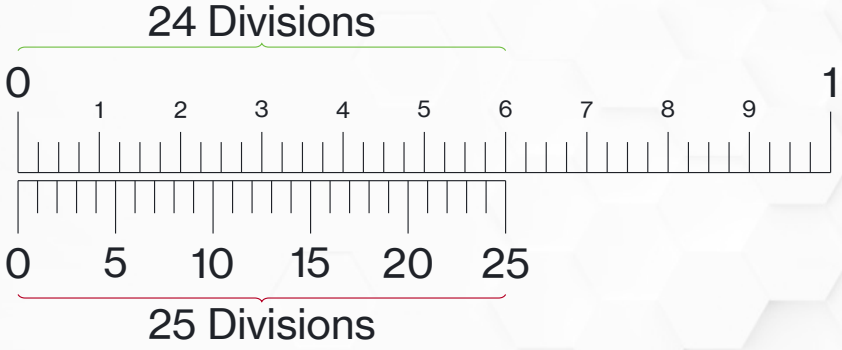
Scale Division Variations

- All vernier scales behave the same way
- Not all vernier scales have the same number of divisions
- Different divisions means different values–almost no real world Vernier uses 1/5

Vernier Fractions

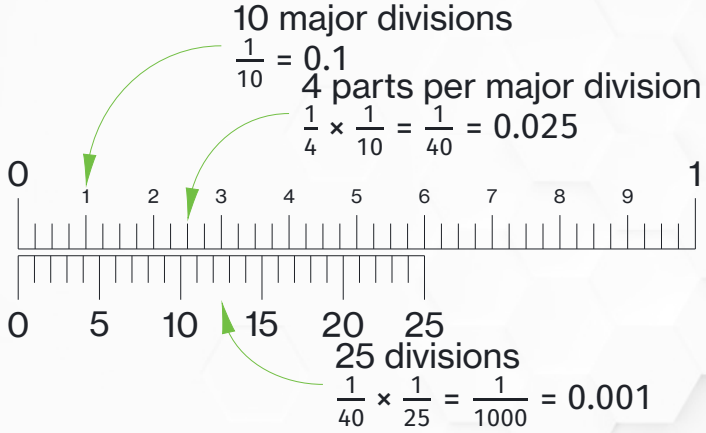
- A Vernier scale is reading a fraction
- The denominator is the number of vernier divisions
- The numerator is the vernier mark that aligns with the main scale
- 10, 25, 50 vernier divisions are common–each of these scales uses a different denominator

25-Division Inch Vernier



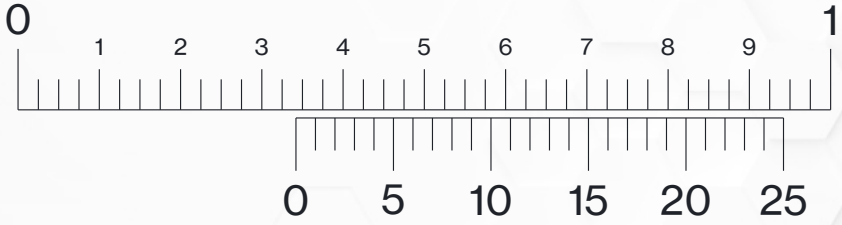
A Vernier that measures in inches typically has 25 divisions.

Inch Scale Precision



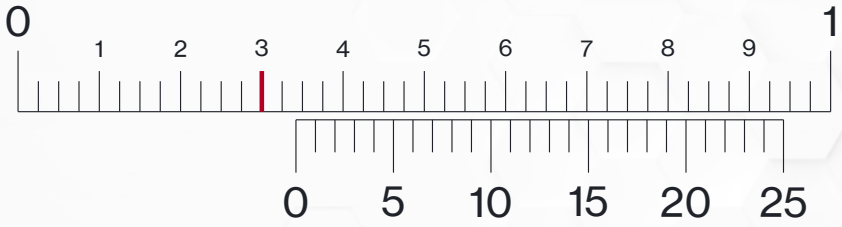
There are 10 major divisions per inch, which means each major division is 0.1 inches. There are 4 parts per major division, which means each part is 0.025 inches. And there are 25 divisions on the Vernier, which means each Vernier division is 0.001 inches.

Example Reading (1 of 4)



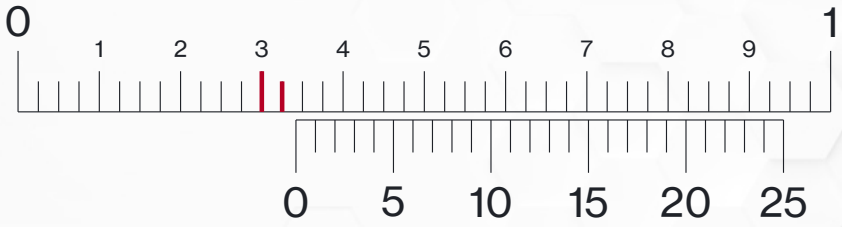
What is the measurement shown in this Vernier?

Example Reading (2 of 4)



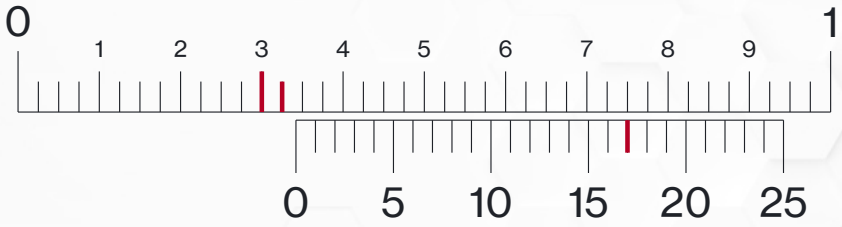
Find the first major division that the 0 on the Vernier passes, then multiply by the fraction that division represents: 3 × 0.1 = 0.3

Example Reading (3 of 4)



Find the first part that the 0 on the Vernier passes, multiply by the fraction that part represents, and add it to the major division: 1 × 0.025 = 0.025, 0.3 + 0.025 = 0.325

Example Reading (4 of 4)

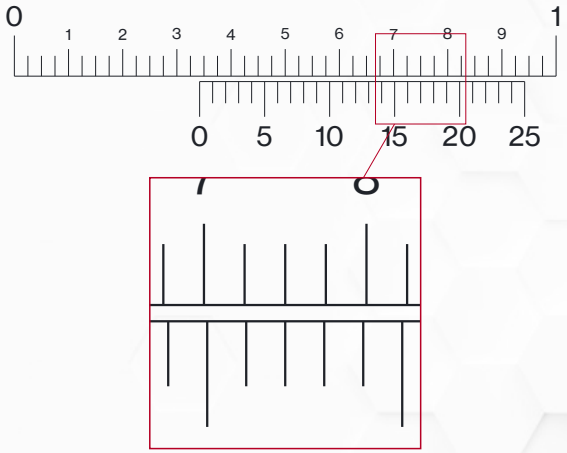


Find the Vernier mark that lines up with *any* mark on the main scale, multiply it by the Vernier fraction, and add it to the total: 17 × 0.001 = 0.017, 0.325 + 0.017 = 0.342

Line Alignment Challenge

- *I* know the 0.017 mark aligns - I drew the picture
- How do *you* know that the 0.017 mark aligns?

Alignment Parallax Error



There will *almost always* be several lines that look like they line up–that's one source of *error* in a Vernier scale.

Best Alignment Practice

- It is always a bit of a guess
- There will usually be 3 marks obviously closer than the others
- Pick the middle one

Video: Metric Calipers

How to Read a Metric Vernier Caliper

https://www.youtube.com/watch?v=vkPlzmalvN4

Video ©WeldNotes via YouTube.com

Micrometers

Common Micrometer Tools

One of the most common measurement tools in machining

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Micrometer Advantages

1. Small, portable, rugged
2. Good measurement range
3. Retains accuracy/wear adjustment
4. Easy to handle/read
5. Inexpensive

Attibution: Mauro Cateb, CC BY-SA 3.0, via Wikimedia Commons

Imperial and Metric Types

- Available in imperial and metric
- Uses the idea of screw *lead* for accurate measurement

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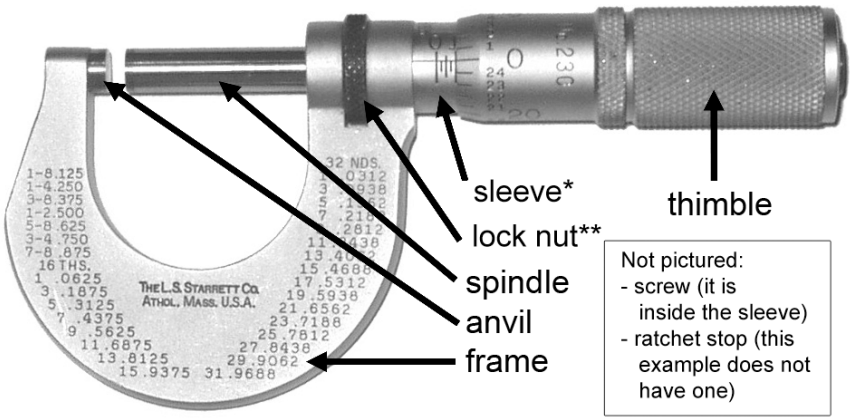
Micrometers

Micrometer Construction

Micrometer Construction

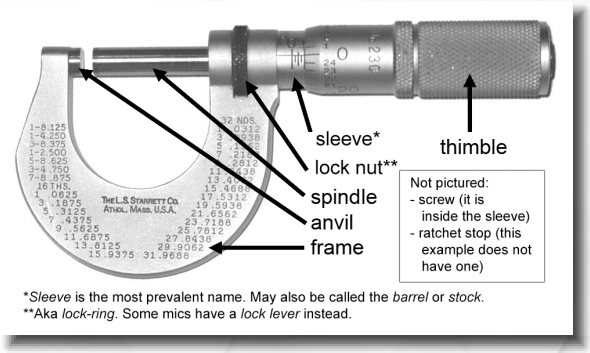


*Sleeve is the most prevalent name. May also be called the *barrel* or *stock*.
**Aka *lock-ring*. Some mics have a *lock lever* instead.

Attibution: Three-quarter-ten, CC BY-SA 4.0, via Wikimedia Commons

Frame and Anvil

- U-shaped frame
- One end: fixed anvil
  - Hardened steel
  - Pressed or screwed into frame



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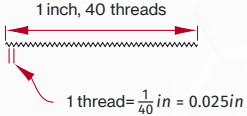
Barrel

- Fixed relative to frame
- Houses the nut that the screw rides in
- Includes a *linear* scale with 0.025” divisions for quick reading

Thimble

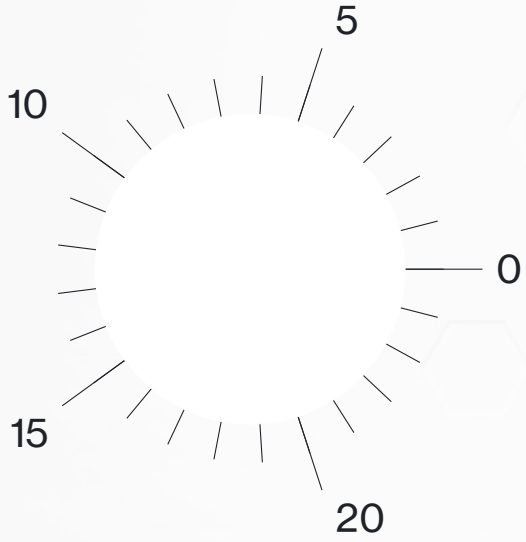
- Used to turn the screw
- Usually knurled for grip
- Includes a scale divided *radially* into 25 equal spaces

Screw



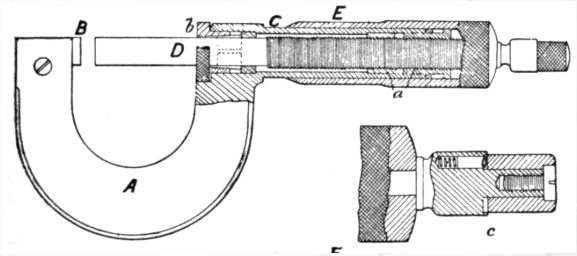
The screw in a micrometer uses custom thread with a *very* precise pitch–the accuracy of the micrometer depends almost entirely on this screw.

The Thimble Scale



Thread Lead Principle

- Internal nut is stationary
- Precision screw – 40 threads per inch
  - Lead = 1/40” = 0.025”
- One revolution = 0.025” axial movement



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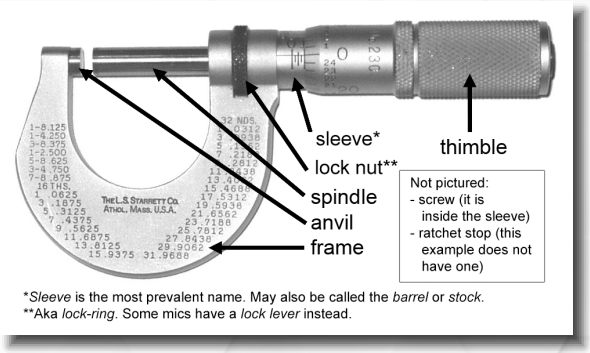
Screw Revolution Reading

- Barrel scale reads full revolutions of the screw
- Thimble scale reads partial revolutions of the screw

Spindle

- Unthreaded part of the screw
- Advances/retracts in the frame
- Supported by a plain bearing in the barrel



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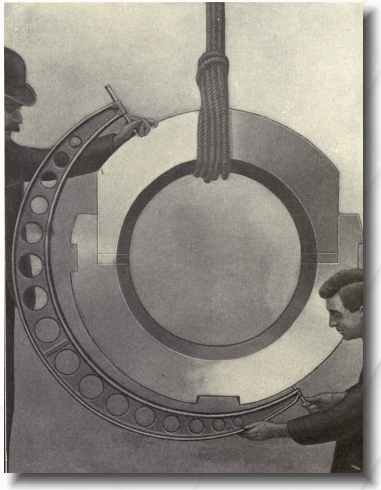
Large Micrometer Offset

- *ALL* imperial micrometers are 1” micrometers
  - A 2” micrometer is still a 1” micrometer
  - Spindle only closes to within 1” of anvil
  - You need to automatically add 1” to all measurements



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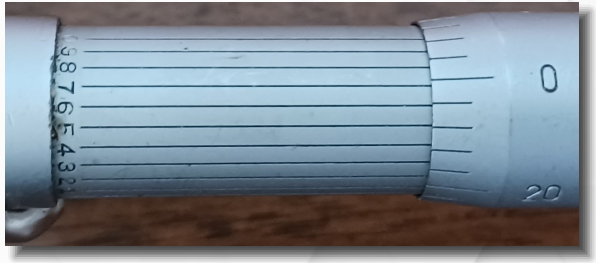
Barrel Vernier Scale

- Some barrels include a *Vernier* scale for 0.0001” resolution
- Top of barrel will be divided into 10 equal spaces



Ratchet

- Measurement pressure is absolutely critical to precision with micrometers
- Many micrometers include a ratchet

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Measuring Pressure

- Thimble used for gross adjustment
- Ratchet used for fine adjustment – slips at correct pressure

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Integrated Ratchet

- May be included as an extension on the back of the thimble
- May be integrated into the thimble itself

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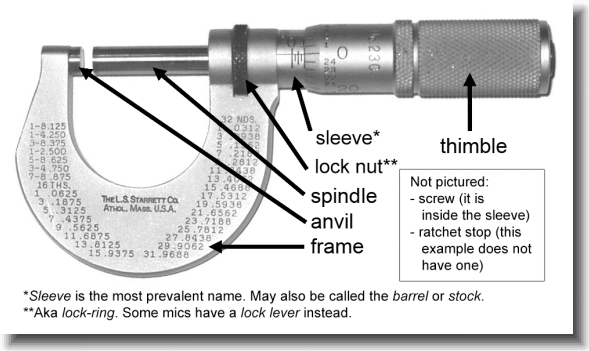
Lock Nut Mechanism



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Micrometers

Reading Micrometers

Barrel Scale Divisions

- The barrel includes a linear scale
- Linear scale is divided into 10 major divisions
- Major divisions = 0.1”

Minor Barrel Divisions

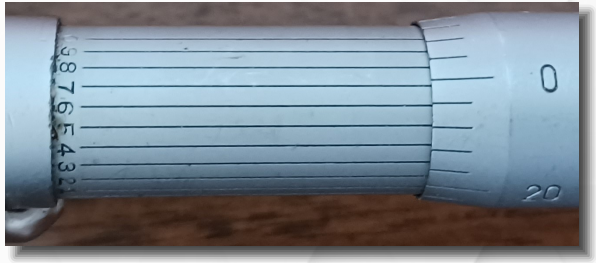
- Each major division is divided into 4 equal spaces
- 0.1” / 4 = 0.025” per minor division
  - One full revolution of thimble = one minor division

Thimble Divisions

- Split into 25 equal divisions, 0.001” apiece

Vernier Resolution

- On top of barrel
- 10 equal division = 0.0001” apiece
- Read like a vernier scale



Part Placement

- Close the part to be measured between the spindle and anvil
  - Remember–pressure matters
  - Use the ratchet if available

Reading the Barrel Scale

- The last visible line is the barrel scale measurement
  - 2 = 0.2”
  - One more visible line = 0.025”
  - Barrel measurement = 0.225”

Reading the Thimble Scale

- The Price is Right–the closest without going over is the thimble measurement
  - 10 is closest to the barrel line
  - 10 = 0.010”

The Total Measurement

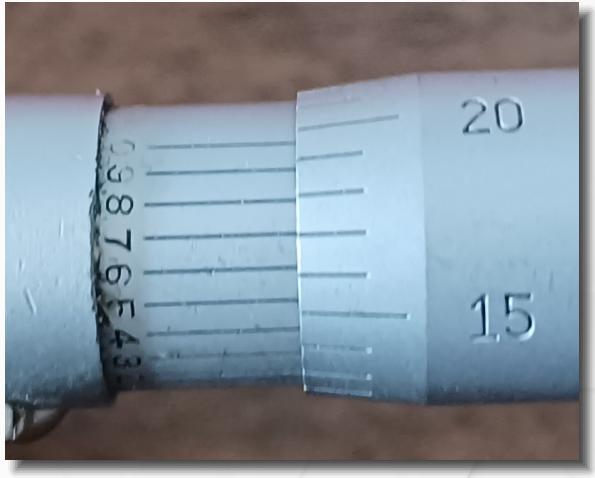
- Total measurement = barrel measurement + thimble measurement
  - 0.225” + 0.010” = 0.235”
- If the micrometer doesn’t include a vernier scale, this is the best that can be measured

Reading the Vernier Scale (1 of 2)

- Notice the measurement is *almost* to 11 on the thimble
- Without a vernier, though, 10 *has* to be the measurement

Reading the Vernier Scale (2 of 2)

- 7 has the best alignment
  - Which barrel line it matches doesn't matter
- 7 = 0.0007”
- 0.235” + 0.0007” = 0.2357”



Video: Metric Micrometer

How to Read a Metric Micrometer by WeldNotes.com

https://www.youtube.com/watch?v=StBc56ZifMs

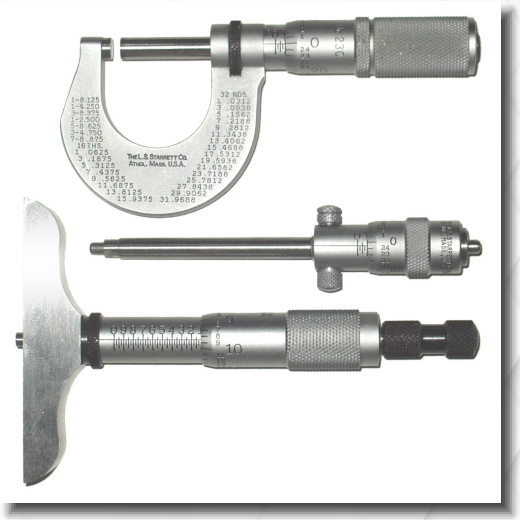
Video ©WeldNotes via YouTube.com

Micrometers

Types of Micrometers

Specialized Micrometers

- Many sizes and styles of micrometers
- Some specialized for specific measurement tasks
- All use the same basic function to perform the measurement



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Outside Micrometers

- The micrometers we've been looking at
- Used to measure outside dimensions of parts
- Available in different sizes
  - Remember: bigger micrometers still only have 1" of stroke

Attibution: Mauro Cateb, CC BY-SA 3.0, via Wikimedia Commons

Telescoping Gauges

- Not a mircometer – commonly used with outside micrometers
- Two expanding rods
- Screw in handle extends/retracts rods
- Gauge expanded to walls of part – transferred to outside micrometer for measurement

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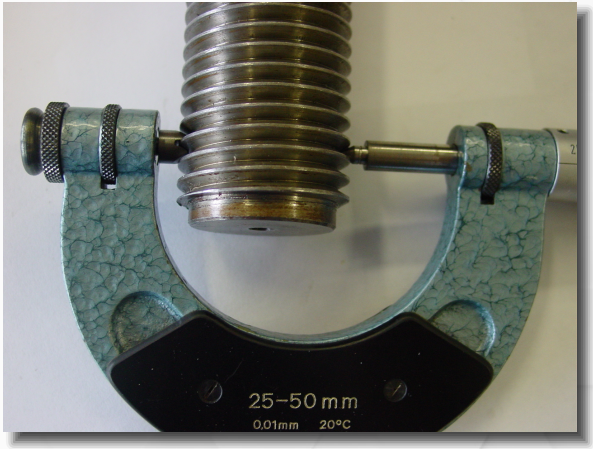
Thread Micrometer

- Special outside micrometer
- Measures screw minor diameter
- Anvil and spindle are pointed to get between screw threads



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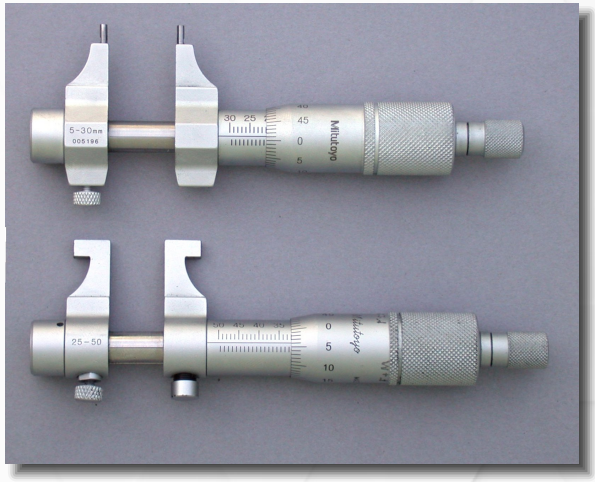
Inside Micrometers

- Used to measure inside dimensions of parts



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Depth Micrometer

- Used to measure the depth of holes/pockets
- Frame is a single, flat surface to rest on part
- No fixed anvil
- Spindle is usually interchangeable for different starting depths

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Questions?

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